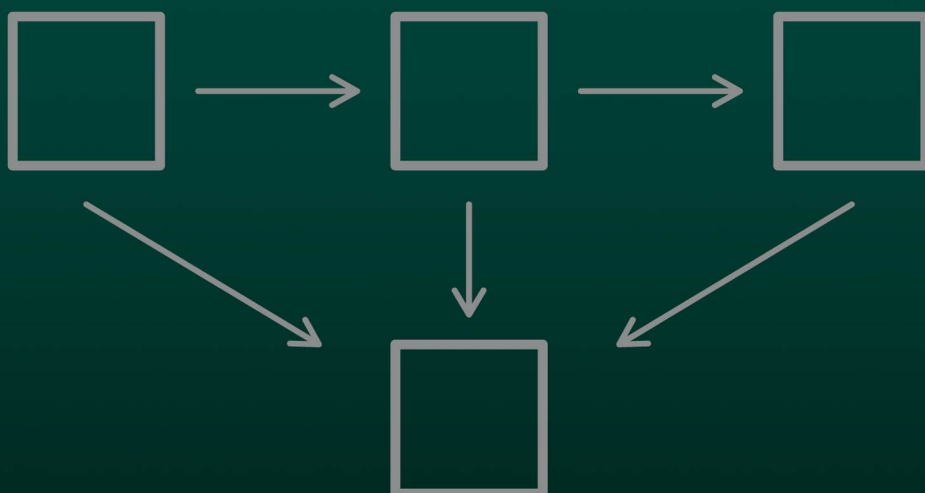


Multi-State Survival Models for Interval-Censored Data



Ardo van den Hout



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Multi-State Survival Models for Interval- Censored Data

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Department of Statistical Science
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Preface

This book is about statistical inference using multi-state survival models. The aim is to introduce and explain methods to describe stochastic processes that consist of transitions between states over time. The book is targeted at applications in medical statistics, epidemiology, demography, and social statistics. An example of an application is a three-state process for dementia and survival in the older population. Such a process can be described by an illness-death model, where state 1 is the dementia-free state, state 2 is the dementia state, and state 3 is the dead state. Statistical analysis can investigate potential associations between the risk of moving to the next state and variables such as age, gender, or education. Statistical analysis can also be used to predict the multi-state survival process. A prediction of specific interest is residual life expectancy; that is, prediction of the expected number of years of life remaining at a given age. When the model describes an illness-death process in the older population, total residual life expectancy at a given age can be subdivided into healthy life expectancy and life expectancy in ill-health.

Applications in this book concern longitudinal data. Typically the data are subject to interval censoring in the sense that some of the transition times are not observed but are known to lie within a given time interval. For example, the time of onset of dementia is latent but when longitudinal data are available the onset may be known to lie in the time interval defined by two successive observations.

Methodologically, multi-state modelling is an elegant combination of statistical inference and the theory of stochastic processes. With this book, I aim to show that the statistical modelling is versatile and allows for a wide range of applications. The computation that is involved in fitting the models can be considerable, but in many cases existing software can be utilised.

I hope that the book will be of interest to diverse groups of readers. Firstly, the book introduces multi-state survival modelling for researchers who are new to the subject. After the introductory first chapter, the book discusses the multi-state survival model as an extension of the (two-state) standard survival model. Further topics are discrete-time models versus continuous-time models, theory on continuous-time Markov chains, parametric models for

transition-specific baseline hazards, maximum likelihood inference, model validation, and prediction. The appendix includes code for the R software.

Secondly, for readers with subject-matter knowledge, the book can serve as a reference and it also offers advanced topics such as the modelling of time-dependent hazards, a general scoring algorithm for maximum likelihood inference, methods for Bayesian inference, estimation of state-specific life expectancies, semi-parametric models, and specification and estimation of frailty models.

For the software used in the data analysis, some details are provided in Appendix C. Software is also available on the author's website, www.ucl.ac.uk/~ucakadl/Book.

The book assumes knowledge of mathematical statistics at third-year undergraduate or at MSc level. This knowledge can be based on courses in statistics or applied mathematics, but also on courses in other disciplines with a strong statistical programme.

Acknowledgments

Thinking about the statistical methods, writing about them, coding the computation, and analysing data have been very much a joint effort in the past years. Discussions with collaborators have helped me to understand both the scope and the finer details of multi-state survival models. Special thanks to Fiona Matthews for supervising my first years of research on this topic, and for all the joint work. I would also like to thank my collaborators Tirza Buter, Jean-Paul Fox, Jutta Gampe, Carol Jagger, Venediktos Kapetanakis, Rinke Klein Entink, Robson Machado, Riccardo Marioni, Graciela Muniz-Terrera, Ekaterina Ogurtsova, Nora Pashayan, Luz Sanchez-Romero, and Brian Tom.

I am grateful to the Medical Research Council Biostatistics Unit in Cambridge, and the Department of Statistical Science, University College London, for allowing me time for research. I would like to thank the colleagues at these two institutes who helped my research with discussions and feedback. Thanks also to those students at University College London who did research projects on multi-state models. These projects have been an additional source for learning and insight.

I am in debt to the people who made it possible to use the longitudinal data for the examples in this book. This includes both the participants in the studies and the teams who collected and managed the data; see Section 1.5.

Thanks to Rob Calver at CRC Press for encouragement to write this book, and for his help and patience. Finally, many thanks to the two anonymous reviewers of the book manuscript. Their detailed and substantial remarks have been a great help in improving the material.



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Introduction

1.1 Multi-state survival models

Multi-state models are routinely used in research where change of status over time is of interest. In epidemiology and medical statistics, for example, the models are used to describe health-related processes over time, where status is defined by a disease or a condition. In social statistics and in demography, the models are used to study processes such as region of residence, work history, or marital status. A multi-state model that includes a dead state is called a *multi-state survival model*.

The specification and estimation of a multi-state survival model depend partly on the study design which generated the longitudinal data that are under investigation. An important distinction is whether or not exact times are observed for transitions between the states. This book considers mainly study designs where death times are known (or right-censored) and where times of transitions between the living states are interval-censored. Many applications in epidemiology and medical statistics have this property as it is often hard to measure the exact time of onset of a disease or condition. Examples are dementia, cognitive decline, disability in old age, and infectious diseases.

Closely related to the interval censoring is the choice between discrete-time and continuous-time models. A discrete-time model assumes a stepwise transition process, where the fixed time between successive steps is not part of the model. There are applications where this model is appropriate, and other applications for which this model is a good approximation of the process of interest. Continuous-time models, the topic of this book, allow changes of state at any time and will be more realistic in many situations. This type of model is also more flexible with respect to the study design for the observation times.

A continuous-time multi-state model can be seen as an extension of the standard survival model. The latter can be defined as a two-state model where a one-off change of status is the event of interest. The archetype example in

medical statistics is the transition from the state of being alive to the state of being dead. Often there will be additional information in the data on other stochastic events that may be associated with the risk of a transition. An example of this is the onset of dementia in a study of survival in the older population. In such a case, the onset of dementia can be taken into account in the survival model by including it as binary time-dependent covariate for the risk of dying. The multi-state model approach would in this case consist of defining three states: a dementia-free state, a dementia state, and a dead state.

Assuming that both models are parametric, there are some clear advantages of the three-state model over the two-state model. Firstly, because the onset of dementia is part of the model, the three-state model can be used for prediction. Even though the two-state model can be used to study the effect of dementia on survival, this model cannot be used for prediction without additional modelling of the stochastic process which underlies the onset of dementia. Secondly, the onset of dementia is a latent process—observation will always be interval-censored. Dealing with this interval censoring is not straightforward when the onset is included in the two-state model as a time-dependent covariate. The multi-state models in this book, however, are explicitly defined for interval-censored transitions between living states.

Even in applications where survival is not of immediate interest, multi-state survival model can still be very useful. Especially in epidemiological and medical research, a longitudinal outcome variable of interest may be correlated with survival. If this is the case, the risk of dying cannot be ignored in the data analysis. Examples of such longitudinal outcomes are blood pressure, cognitive function in the older population, and biomarkers for cardiovascular disease. One option is to specify a model for the longitudinal process of interest and combine this model with a standard survival model. This is called a joint model, and it is often specified with random effects which are shared by the two constituent models. But if the longitudinal outcome can be adequately discretised by a set of living states, then a multi-state survival model can be defined by adding a dead state. This provides an elegant alternative to a joint model: the multi-state model can be defined as an overall fixed-effects model for the process of interest as well as for survival.

A multi-state process is called a *Markov chain* if all information about the future is contained in the present state. If, for example, time spent in the present state affects the risk of a transition to the next state, then the process is not a Markov chain. Although many processes in real life are not Markovian, a statistical model based upon a Markov chain may still provide a good approximation of the process of interest. Most of the multi-state models in this book are not Markovian in the strict sense. By linking age and values of

covariates to the risk of a transition, information about the future is contained not only in the present state, but also in current age and additional background characteristics.

This first chapter introduces the basic concepts that are used in multi-state survival models, discusses the relevant type of data, and illustrates the scope of the statistical modelling by an example. Details of parameter estimation and statistical inference are postponed to later chapters. At the end of this chapter an overview will be given of the methods and the examples in the rest of the book.

1.2 Basic concepts

The standard two-state survival model is defined by distinguishing a living state and dead state. The two main features of the standard survival model are that there is one event of interest (the transition from alive to dead) and that the timing of this event may be right-censored, in which case it is known that the event has not happened yet.

As an example, say patients are followed up for a year after a risky medical operation and the time scale is months since the operation. When the event is defined by death, the information per patient is either time of death or the time at which death is right-censored. The latter is typically the end of the follow-up, in this example 12 months. Alternatively, the patient drops out of the study during the follow-up and the censored time is the last time the patient was seen alive. The statistical modelling of survival in the presence of right censoring is often undertaken by assuming a model for the hazard. The hazard is the instantaneous risk of the event. As a quantity it is an unbounded positive value and should be distinguished from a probability, which is a value between zero and one. In the example, the probability of the event is linked to a specified time interval, for example, the probability of dying within a year, whereas the hazard is defined for a moment in time specified in months.

In a multi-state survival model there is more than one event of interest. An example with two states is the functioning of a machine with working (state 1) versus being repaired (state 2). The events are the transition from state 1 to state 2 and the transition from state 2 to state 1. An example with three states is an illness-death model where state 1 is the dementia-free state, state 2 is the state with dementia, and state 3 is the dead state. This defines three events: death from state 1, death from state 2, and onset of dementia.

A multi-state survival process can be described by a model for the transition-specific hazards. There are as many hazards to model as there are transitions. Censoring is often more complex than in the standard survival

model. In the example with the dementia state, for an individual who is last seen in state 1, right censoring at a later time concerns both death and a potential transition to the dementia state. Furthermore, often it is known that a transition between states took place without knowing the exact time of the transition. This is called *interval censoring*. The onset of dementia between two screening times is an example of this.

For the formal definition of interval censoring, the following summarises the discussion in Sun (2006). Let random variable T denote the time to the event. The sample space of T is $[0, \infty)$. An observation on T is interval-censored if only an interval $(L, R]$ is observed such that

$$T \in (L, R].$$

This notation extends to exact observations of T in case $L = R$, and to right-censored observations in case $R = \infty$. Interval-censored data in this book refer to the situation where the data include at least one interval $(L, R]$ with $L \neq R$ and both L and R in $(0, \infty)$. In the case of longitudinal data where multiple events are of interest, interval-censored data are also called *panel data* (Kalbfleisch and Lawless, 1985), or *intermittently observed data* (Tom and Farewell, 2011; Cook and Lawless, 2014).

Independent interval censoring implies that the mechanism that generates the censoring is independent of T . An example of this for longitudinal data is when all observation times during follow-up are determined in advance at baseline. Formally, the interval censoring is independent when

$$P(L < T \leq R | L = l, R = r) = P(l < T \leq r),$$

that is, when the joint distribution of L and R is free of the parameters involved in the distribution of T (Sun, 2006, Section 1.3). If the interval censoring is independent, then the censoring mechanism can be ignored in the data analysis.

In biostatistics, complete independence between observation times and a multi-state disease process is often not realistic. Given random variable M for the number of observations, Grüger et al. (1991) define an observation scheme $\{T_1, \dots, T_m | M = m\}$ to be *non-informative* for the multi-state disease process if the full likelihood on $\{T_1, \dots, T_m | M = m\}$ is proportional to the likelihood obtained if the number of observations and their times were fixed in advance. It follows that if the observation scheme is non-informative, then the scheme can be ignored when maximum likelihood is used to estimate parameters for the multi-state process.

Independent interval censoring implies a non-informative observation scheme but not vice versa. An important and illustrative example of an

observation scheme that induces dependent interval censoring but is non-informative is *Doctor's care* as defined in Grüger et al. (1991). In this scheme, a next observation time is chosen on the basis of the current state. If states are defined with respect to disease progression, this would imply shorter time intervals between successive observations for patients with increased risk of dying, and longer intervals for patients in the healthy state or in a stable stage of the disease.

An example of an observation scheme that is not non-informative is patient self-selection. In this scenario, patients go to the doctor because of their disease progression. If such a visit is included as an observation of the state in the longitudinal data, then this leads to biased likelihood inference when the self-selection is not taken into account.

1.3 Example

1.3.1 Cardiac allograft vasculopathy (CAV) study

To illustrate the statistical modelling of a multi-state survival process, we use data with individual histories of angiographic examinations of 622 heart transplant recipients. Data were collected at Papworth Hospital in the United Kingdom and are included in the R package *msm* (Jackson, 2011). The data are available for illustration purposes only and should not be used in isolation to inform clinical practice. Permission to use the data for this book was kindly given by Steven Tsui at Papworth Hospital.

Cardiac allograft vasculopathy (CAV) is a deterioration of the arterial walls. For the heart transplant recipients, four states are defined: no CAV, mild/moderate CAV, severe CAV, and dead; see Sharples et al. (2003). Figure 1.1 depicts the states and transitions in the data.

Yearly examinations for CAV took place after the transplant for up to 18 years. There is variation in time between examinations: it was not always exactly 1 year and some patients skipped one or more scheduled examinations. For example, the patient who was followed up for 18 years and whose death is right-censored has observed states at 0, 1.01, 2.01, 3.01, 4.07, 5.00, 6.01, 7.92, 8.92, 11.03, 12.01, 15.63, and 17.98 years after transplant. For the whole sample, the follow-up in years until right censoring or death is given in Figure 1.2. The histogram shows that a follow-up with observation during the whole period of 18 years is exceptional. The mean of the follow-up time is 5.9 years and the median is 5.0 years.

Transitions between the living states are interval-censored, but if the patient died during follow-up, then his or her death time is known exactly.

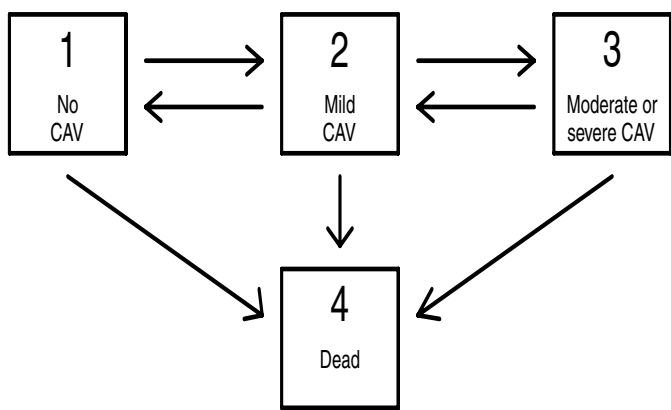


Figure 1.1 *Transitions in the data for the four-state model for cardiac allograft vasculopathy (CAV).*

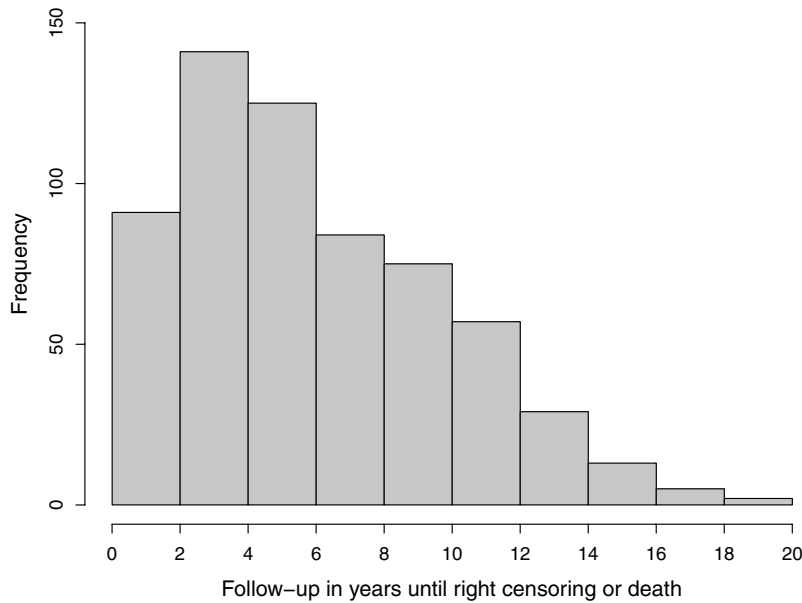


Figure 1.2 *For the CAV data, frequencies of follow-up time in years until right censoring or death.*

Figure 1.3 shows how observed state changes over time. The sizes of the diameters correspond with the observed frequency in the corresponding state. Due to the interval censoring, the information in Figure 1.3 is limited. For example, the state of the individual who was followed up for 18 years

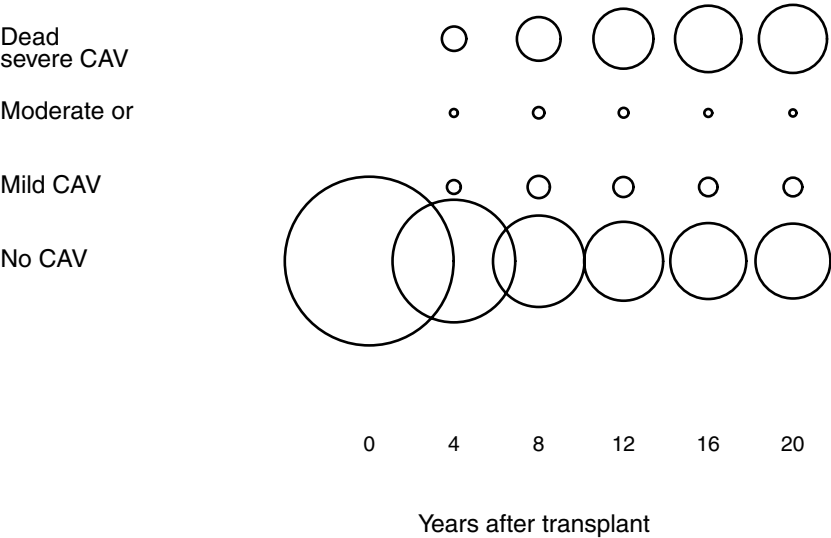


Figure 1.3 For the CAV data, last seen state for an imposed time grid in years after transplant. Difference in diameters refers to difference in frequencies. All $N = 622$ individuals start in the state without CAV.

and whose observation times are given above was not observed at year 16. At 15.63 years after transplant, he or she did not have CAV, but at 17.98 years mild CAV was observed. Whether this individual obtained CAV before 16 years after transplant or later is not known. When we do not have exact transition times, we only know the state that was observed at the last time an individual was seen. Hence, the diagram is only an approximation of the true status of the process at the specified time points.

Despite the fact that the diagram in Figure 1.3 is an approximation it gives a good idea of how the process develops over time. It shows that all individuals start without CAV and that most of the movement between the states takes place in the first 12 years. Note that the diagram cannot distinguish whether the states with CAV are relatively rare or that the duration in these states is short. The dead state is absorbing and the frequency for that state does not decrease. The frequency for the no-CAV state does not increase over time, which agrees with the general idea of CAV being progressive.

Further information about the CAV process is given by the frequencies in Table 1.1. This is called the *state table* and is a way to summarise multi-state data. The frequencies are the number of times each pair of states was observed at successive observation times. For the CAV data, the diagonal for the living states dominates which illustrates that the process of change is slow relative

Table 1.1 *State table for the CAV data: number of times each pair of states was observed at successive observation times.*

From	To			
	No CAV	Mild CAV	Mod./Severe CAV	Dead
No CAV	1367	204	44	148
Mild CAV	46	134	54	48
Mod./severe CAV	4	13	107	55

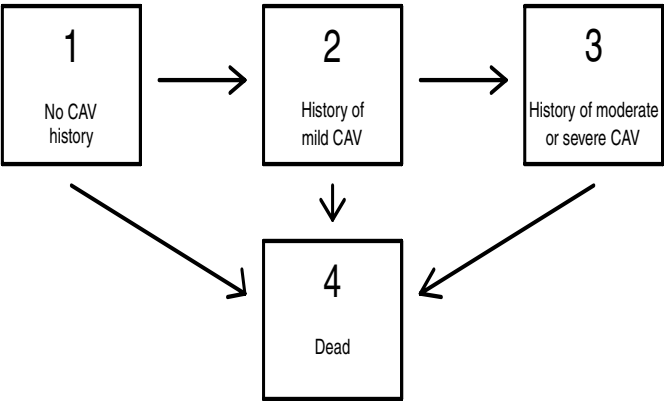


Figure 1.4 *Four-state progressive model for history of cardiac allograft vasculopathy (CAV).*

to the timing of the follow-up. Note that from the moderate/severe state there are just a few observed backward transitions, which shows the progressive nature of the deterioration of the arterial walls.

1.3.2 A four-state progressive model

Sharples et al. (2003) assume that CAV is a progressive process where recovery is not possible and that observed recoveries are due to misclassification of states. Multi-state models have been developed for this situation. Sharples et al. (2003) use such a model to analyse the CAV data by distinguishing a latent progressive multi-state process from the observed manifest process. More information on how misclassification can be taken into account in multi-state survival models is given in Section 8.5, which includes an example with the CAV data.

For the current analysis, we define a progressive multi-state model for observed history of CAV. The time scale t in the model is *time in years since transplant*. CAV history at time t is defined as the highest CAV state observed up to and including t . This implies that there are five transitions; see Figure 1.4. States 1, 2, and 3 are defined as the states without CAV history, with a history of mild CAV only, and with a history of moderate/severe CAV, respectively. State 4 is the dead state.

For each of the five transitions the hazard is modelled using log-linear regression equations. In these equations, the covariates are time since transplant (t), baseline age at transplant ($b.age$), and age of the donor ($d.age$). The hazard for the transition from state r to state s at time t is denoted by $q_{rs}(t)$, and the regression equations are given by

$$\begin{aligned}
 q_{12}(t) &= \exp(\beta_{12.0} + \beta_{12.1}t + \beta_{12.2} b.age + \beta_{12.3} d.age) \\
 q_{14}(t) &= \exp(\beta_{14.0} + \beta_{14.1}t + \beta_{14.2} b.age + \beta_{14.3} d.age) \\
 q_{23}(t) &= \exp(\beta_{23.0} + \beta_{23.3} d.age) \\
 q_{24}(t) &= \exp(\beta_{24.0} + \beta_{24.3} d.age) \\
 q_{34}(t) &= \exp(\beta_{34.0} + \beta_{34.3} d.age).
 \end{aligned} \tag{1.1}$$

Many more models can be defined for the CAV-history data. Specification (1.1) is used as an example. It reflects our interest in the age of the donor as a risk factor for leaving state 1. To control for the effects of time since transplant and baseline age, these two covariates are also included in the relevant regression equations.

Maximum likelihood can be used to estimate the model parameters. The details for defining the likelihood function are presented in Section 4.5. An important feature of the statistical inference for (1.1) is that piecewise-constant hazards are used in the maximum likelihood estimation. The piecewise-constant hazards are an approximation of the continuous-time hazard specification in (1.1).

Grid points for the approximation can be defined by the data; that is, the grid in the likelihood contribution for a given individual is constructed using individual observation times. Given the survey design of yearly observations, this implies in most cases that the approximation of the hazard allows a step-wise change from year to year. If an individual skips one planned observation, the approximation is cruder and the hazard stays constant for 2 years. For example, the likelihood contribution for the individual who was mentioned above and who was followed up for 18 years is defined using constant hazards within the intervals $(0, 1.01]$, $(1.01, 2.01]$, $(2.01, 3.01]$, etc.

Table 1.2 *Parameter estimates for the four-state model for the CAV-history data. Estimated standard errors in parentheses. Intercepts, and effects of time since transplant (t), age at transplant ($b.age$), and age of the donor ($d.age$).*

Intercept		t		$d.age$	
$\beta_{12.0}$	-3.476 (0.356)	$\beta_{12.1}$	0.118 (0.025)	$\beta_{12.3}$	0.026 (0.006)
$\beta_{14.0}$	-6.427 (0.724)	$\beta_{14.1}$	0.019 (0.051)	$\beta_{14.3}$	0.023 (0.010)
$\beta_{23.0}$	-1.250 (0.297)			$\beta_{23.3}$	-0.006 (0.009)
$\beta_{24.0}$	-1.896 (1.194)	$b.age$		$\beta_{24.3}$	-0.042 (0.048)
$\beta_{34.0}$	-1.057 (0.388)	$\beta_{12.2}$	0.002 (0.007)	$\beta_{34.3}$	-0.007 (0.013)
		$\beta_{14.2}$	0.052 (0.012)		

Using the piecewise-constant approximation with a grid defined by the data implies that the estimation of the model parameters becomes data-dependent. The model itself still assumes a continuous-time dependency. This can be distinguished from a piecewise-constant hazard model which explicitly assumes that the hazard is constant from one interval to the next.

The main advantage of the piecewise-constant hazard approximation is that it simplifies the estimation of the model parameters and—in this case—that the maximum likelihood estimation can be undertaken using the R package *msm* (Jackson, 2011). Parameters can also be estimated using the scoring algorithm for multi-state survival models presented in Chapter 4, or using the sampling methods for Bayesian inference in Chapter 6.

Table 1.2 presents the model parameters in (1.1) as estimated by maximum likelihood where the piecewise-constant approximation is defined by the data. An absolute value for the hazard is of limited information on its own, but values can be compared. Table 1.2 shows that with respect to covariate effects, an increase of years since transplant is associated with a higher risk of moving from the state without CAV history to the state with a history of moderate/severe CAV. Regarding the age of the donor, the estimate of the regression coefficient $\beta_{12.3}$ in Table 1.2 shows that a higher donor age is associated with an increase risk of moving out of the state without CAV history. For the states with CAV history, the age of the donor does not seem to have a significant effect.

More informative than hazards are transition probabilities, which are the probabilities of transition between states within a specified time interval. It is common to present these probabilities in a matrix. In the current application, this is a 4×4 matrix, where the rows are the current state and the columns the next state. Assume we are interested in the first year after the transplant.

At the time of the transplant, median age of the patients in the sample is 49.5 years. Median donor age is 28.5. Conditional on these median values, we obtain the probability matrix

$$\begin{aligned}\hat{\mathbf{P}}(t) &= \begin{pmatrix} \hat{p}_{11}(t) & \hat{p}_{12}(t) & \hat{p}_{13}(t) & \hat{p}_{14}(t) \\ \hat{p}_{21}(t) & \hat{p}_{22}(t) & \hat{p}_{23}(t) & \hat{p}_{24}(t) \\ \hat{p}_{31}(t) & \hat{p}_{32}(t) & \hat{p}_{33}(t) & \hat{p}_{34}(t) \\ \hat{p}_{41}(t) & \hat{p}_{42}(t) & \hat{p}_{43}(t) & \hat{p}_{44}(t) \end{pmatrix} \\ &= \begin{pmatrix} 0.894 & 0.060 & 0.007 & 0.039 \\ 0 & 0.752 & 0.181 & 0.067 \\ 0 & 0 & 0.755 & 0.245 \\ 0 & 0 & 0 & 1 \end{pmatrix},\end{aligned}$$

where $t = 1$ year. For the first three elements of the diagonal, the 95% confidence intervals are (0.877, 0.908), (0.707, 0.783), and (0.707, 0.800), respectively. For example, the entry $\hat{p}_{12}(t)$ implies that for someone who is 49.5 years old at the time of transplant and who gets a donor heart from a 28.5 year old has an estimated probability of 0.060 of having a history of mild CAV after 1 year.

As an alternative to the above piecewise-constant approximation where individual data determine the grid for the approximation in the likelihood contribution per individual, a fixed grid can be used that is imposed for all likelihood contributions. This is still an approximation of the time-continuous change of the hazard in (1.1), but there are two advantages. Firstly, the grid is the same across individuals and independent of the data. Secondly, if there are individuals with long time intervals between two successive observations, the imposed grid subdivides those long intervals into shorter ones resulting in a better approximation of the continuous-time dependency.

The choice between using individual grids or an imposed grid can make a difference in the estimation of model parameters. This will depend on the study design in combination with the volatility of the process of interest.

For the CAV data, we fitted the model defined by (1.1) using an imposed grid with time points 0, 2, 4, 6, 10, and 15 years after baseline. Results were similar compared to the previous analysis. For example, the estimated transition matrix for the first year conditional on median age at the time of transplant and median donor age is

$$\hat{\mathbf{P}}(t) = \begin{pmatrix} 0.896 & 0.059 & 0.007 & 0.039 \\ 0 & 0.750 & 0.181 & 0.069 \\ 0 & 0 & 0.759 & 0.241 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

There are alternatives to using a piecewise-constant approximation for estimation of time-dependent models. Chapter 3 discusses using numerical integration for progressive three-state models. Yet another option is presented by Titman (2011) who uses direct numerical solutions to the Kolmogorov Forward equations which are the differential equations that define the multi-state transition probabilities.

Further understanding of the multi-state process can be derived from estimating state-specific length of stay. Given that the model contains a dead state, length of stay can be formulated at residual state-specific life expectancy. In the CAV context, it is of interest how remaining total life expectancy subdivides into life expectancies in the living states 1, 2, and 3, and whether there is heterogeneity depending on covariate values at the time of transplant. Life expectancies can be derived from a fitted model using transition probabilities. This will be discussed in Chapter 7.

The hazard specification in (1.1) is just one of the many ways a continuous-time hazard model can be defined. If we write the first equation in (1.1) as

$$q_{12}(t) = \lambda_{12} \exp(\xi_{12}t) \exp(\beta_{12.2} \text{ b.age} + \beta_{12.3} \text{ d.age}),$$

it becomes apparent that a Gompertz baseline hazard is used for time t with parameters $\lambda_{12} > 0$ and ξ_{12} . An alternative would be to use a Weibull baseline hazard specified by

$$q_{12}(t) = \lambda_{12} \tau_{12} t^{\tau_{12}-1} \exp(\beta_{12.2} \text{ b.age} + \beta_{12.3} \text{ d.age}),$$

for $\lambda_{12}, \tau_{12} > 0$. These specifications and others will be discussed in Chapters 2, 3, and 4.

1.4 Overview of methods and literature

The multi-state survival models in this book are based upon the theory on *stochastic processes*. In probability theory, a stochastic process is a set of random variables representing the evolution of a process over time. More formally, the process is a collection of random variables $\{Y_t | t \in \mathcal{U}\}$, where $\mathcal{U}$ is some index set. If $\mathcal{U} = \{0, 1, \dots\}$, then the process is in discrete time. If $\mathcal{U} = (0, \infty)$, then the process is in continuous time. Variable Y_t is the *state* of the process at time t . The theory on basic stochastic processes is well established and can be found in many textbooks; see, for example, Cox and Miller (1965), Norris (1997), Ross (2010), and Kulkarni (2011).

In probability theory, probabilistic properties are defined and are used to derive mathematical insight in the behaviour of a process over time. Statistics is primarily an applied branch of mathematics that uses definitions and results from probability theory to describe, understand, and predict processes in the material world. To be able to do this, statistics needs data and methods for distributional inference.

This book is concerned with the statistical modelling and estimation of multi-state processes. The most basic multi-state survival process is the standard survival model which consists of two states, alive and dead, and where there is only one event of interest, namely, death. Textbooks on this topic are, for example, Cox and Oakes (1984), Kalbfleisch and Prentice (2002), Collett (2003), and Aalen et al. (2008). The survival model will serve in Chapter 2 as a starting point for the statistical methods for more complex multi-state processes in Chapters 3 and 4. In multi-state models other than the standard survival model there is more than one event of interest. In a two-state model, for example, it is possible that not only the event of moving from state 1 to state 2 is of interest, but also the event of a transition back to state 1. Many processes in the material world can be described by multi-state models. The focus of this book is on biostatistics and on interval-censored data for continuous-time processes which include one absorbing state, typically the dead state.

Important methodological work on continuous-time multi-state models in the presence of interval censoring is presented in Kalbfleisch and Lawless (1985), Kay (1986), and Satten and Longini (1996). Jackson (2011) presents the freely available R package *msm* that provides a flexible framework for fitting a wide range of continuous-time multi-state models. Many textbooks on survival analysis contain extended material on multi-state models; see, for example, Hougaard (2000), Aalen et al. (2008), and Crowder (2012). The journal *Statistical Methods in Medical Research* devoted a whole issue to multi-state models; see Andersen (2002).

This book provides an overview of methods for multi-state survival models for interval-censored data, and extends the current methods by exploring and estimating time-dependent hazard models for transition intensities. A very general scoring algorithm for maximum likelihood estimation is presented in Section 4.6. Some of the material in this book is based on published work; see, for example, Van den Hout and Matthews (2008), Van den Hout and Matthews (2009), Van den Hout and Matthews (2010), Van den Hout and Tom (2013), and Van den Hout et al. (2014).

There are also topics related to multi-state processes which will not be discussed in depth in this book. One of these topics is the case where there is no interval censoring and exact times are available for all transitions in a

continuous-time multi-state process. For modelling this kind of data, methods based upon counting processes can be used; see, for example, Aalen et al. (2008) and Putter et al. (2007) for a review of these methods and further references.

Attention to discrete-time multi-state processes is also limited. Section 4.1 defines these processes formally, but mainly as a contrast to continuous-time processes. In Section 8.1, the discrete-time process is used to approximate a continuous-time process.

For the longitudinal data, we assume that the sampling is independent of the history of the events of interest. Typically, this implies a prospective study where a random sample is taken from the population and observed subsequently. This independence assumption does not hold for retrospective studies. For the latter, see Kalbfleisch and Lawless (1988) for details and methods.

Two recent books on multi-state models are Beyersmann et al. (2012) and Willekens (2014). The former is on methods for longitudinal data with known transition times—the analysis of interval-censored data is not covered explicitly. The latter discusses interval-censored data to some extent using the term *status data*, and has a strong focus on how to use existing software for multi-state models. Unavoidably, there is some overlap between these two books and the current book. Specific for the current book, however, is the focus on interval-censored data, and the aim for a wider scope of the statistical modelling. This aim is realised by using a very general definition of the multi-state model which is unrestricted with respect to the number of states *and* with respect to possible (forward and backward) transitions. In addition, the current book discusses methods for Bayesian inference and frailty models, and includes a number of further topics of interest such as misclassification of state and missing data.

1.5 Data used in this book

The following is an overview of the data used in this book. Data are used for illustration purposes and the statistical analyses in this book should not be used in isolation to inform clinical practice.

- The cardiac allograft vasculopathy (CAV) data were collected at Papworth Hospital in the United Kingdom and are included in the R package *msm* (Jackson, 2011). Permission to use the data was given by Steven Tsui at Papworth Hospital. The data are introduced in Section 1.3.1. For up to 18 years, annual examinations for CAV took place after heart transplant. There is variation in time between examinations: it was not always exactly 1 year and some patients skipped one or more scheduled examinations. States

are defined according to CAV status and death, and transitions between the living states are interval-censored. The number of individuals in the sample is 622.

- The Parkinson's disease data are provided by Dag Aarsland at the Norwegian Centre for Movement Disorders, Stavanger University Hospital, Norway. The data are introduced in Section 3.7.1. Individuals with Parkinson's disease were followed up between 1993 and 2005. Of interest are survival and the onset of dementia, which is interval-censored or right-censored. The number of individuals in the sample is 233.
- Longitudinal data from the English Longitudinal Study of Ageing (ELSA) are introduced in Section 4.8.1. The ELSA baseline is a representative sample of the English population aged 50 and older. ELSA contains information on health, economic position, and quality of life. Data from ELSA can be obtained via the UK Economic and Social Data Service. Follow-up data are used from 1998 up to 2009. The total number of individuals at baseline is 19,834. This book uses a random subsample of 1,000 individuals to illustrate the methods.
- The Medical Research Council Cognitive Function and Ageing Study (MRC CFAS) is a population-based longitudinal study of cognition and health conducted between 1991 and 2004 in the older population of England and Wales. The data are introduced in Section 4.10.1. Data were collected in two rural areas and three cities (Cambridgeshire, Nottingham, Gwynedd, Newcastle, and Oxford, respectively), with a sample size of at least 2,500 individuals at each site. In this book, the data from Newcastle are used with 2,512 individuals at baseline.
- Dutch data on body mass index (BMI) are used in Section 8.2.1. The data are cross-sectional, and men in the age range 15 up to 40 years old are classified in three states according to their BMI. Permission to use the data for this book was given by Jan van de Kastele at the National Institute for Public Health and the Environment, RIVM, in Bilthoven, the Netherlands. The data used in this book are a subset of the data published in Van de Kastele et al. (2012).



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